

# Gauge Coupling Boundary Conditions from the Unified Mechanics Axiom

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## Abstract

The single recursion  $c^2 = c + 1$  determines the contraction rate  $r = 1/(2\varphi)$  and, with no additional free parameters, derives all three Standard Model gauge couplings at the  $Z$ -boson scale. This is the first instance in which a single axiom fixes the complete gauge-coupling boundary condition:  $\alpha_{\text{em}}$ ,  $\sin^2\theta_W$ , and  $\alpha_s$  simultaneously. The SM 1-loop running produces pairwise intersections spanning  $\mu = 10^{14}$ – $10^{17}$  GeV; the UM-natural grand unification scale  $M_{\text{Pl}} \varphi^{-14} \approx 1.45 \times 10^{16}$  GeV lies within this band, consistent with the MSSM value  $\sim 2 \times 10^{16}$  GeV. Full unification requires the physics of the second  $E_8$  sector, identified as the subject of a companion heterotic paper.

## 1 Foundations

The axiom and its immediate consequences:

$$c^2 = c + 1 \quad \Rightarrow \quad \varphi = \frac{1+\sqrt{5}}{2}, \quad r = \frac{1}{2\varphi} \approx 0.30902. \quad (1)$$

The three channel weights and the braiding floor follow algebraically:

$$W_L = (1 - r)^2 \approx 0.4775, \quad (2)$$

$$W_B = 2r(1 - r) \approx 0.4271, \quad (3)$$

$$W_M = r^2 \approx 0.0955, \quad (4)$$

$$\varepsilon_{\text{floor}} = r^3 \approx 0.02951. \quad (5)$$

$W_L + W_B + W_M = 1$  by construction, as required for a probability partition. Every result in this paper follows from (??)–(??) alone.

A residual is said to be *sub-floor* when its magnitude is less than  $\varepsilon_{\text{floor}} \approx 2.95\%$ , meaning no additional correction cycle is geometrically mandated. A residual between one and a few floors indicates a correction derivable from the next order of channel braiding.

## 2 The Three Gauge Couplings at $M_Z$

### 2.1 The electromagnetic coupling

Paper 1 of this series establishes the Fibonacci cycle-sum identity for the fine structure constant:

$$\frac{1}{\alpha_{\text{em}}} = \varphi^{10} + \varphi^5 + \varphi^2 + \varphi^{-2} = 137.082, \quad (6)$$

giving  $\alpha_{\text{em}} = 0.007295$ . The observed value is  $1/\alpha_{\text{em}} = 137.036$ ; the residual is  $+0.034\%$ , or  $0.01 \varepsilon_{\text{floor}}$ . The exponents  $\{10, 5, 2, -2\}$  match the EM-sector winding numbers in the  $G_2$  compactification; their full derivation is reserved for the companion heterotic paper.

### 2.2 The Weinberg angle and the $\text{SU}(2)_L$ coupling

Paper 2 derives the electroweak mixing angle from the ratio of channel weights:

$$\sin^2 \theta_W = \frac{W_M}{W_B} = \frac{r}{2(1-r)} = 0.22361. \quad (7)$$

The on-shell measured value is 0.22290 (PDG 2022); the residual is  $+0.32\% = 0.11 \varepsilon_{\text{floor}}$ .

At tree level,  $\sin^2 \theta_W = \alpha_{\text{em}}/\alpha_2$ , so the  $\text{SU}(2)_L$  coupling is determined:

$$\alpha_2 = \frac{\alpha_{\text{em}}}{\sin^2 \theta_W} = \frac{2(1-r)}{r} \alpha_{\text{em}} = 0.032628, \quad \frac{1}{\alpha_2} = 30.653. \quad (8)$$

The GUT-normalised  $\text{U}(1)$  hypercharge coupling is

$$\alpha_1 = \frac{5}{3} \frac{\alpha_{\text{em}}}{1 - \sin^2 \theta_W} = 0.015665, \quad \frac{1}{\alpha_1} = 63.858, \quad (9)$$

where the factor  $5/3$  is the standard  $\text{SU}(5)$  hypercharge normalisation.

### 2.3 The strong coupling

Paper 5 derives the strong coupling from the colour-channel geometry:

$$\alpha_s(M_Z) = r^2(1 + r - r^2) = 0.11591, \quad \frac{1}{\alpha_s} = 8.629. \quad (10)$$

The world-average value is  $\alpha_s(M_Z) = 0.11790$  (PDG 2022); the residual is  $-1.69\% = 0.57 \varepsilon_{\text{floor}}$ , sub-floor.

## 2.4 Summary of boundary conditions

| Coupling               | UM expression                                         | UM value | Observed    | Error  |
|------------------------|-------------------------------------------------------|----------|-------------|--------|
| $1/\alpha_{\text{em}}$ | $\varphi^{10} + \varphi^5 + \varphi^2 + \varphi^{-2}$ | 137.082  | 137.036     | +0.03% |
| $\sin^2\theta_W$       | $r/[2(1-r)]$                                          | 0.22361  | 0.22290     | +0.32% |
| $1/\alpha_2$           | $\alpha_{\text{em}}/\sin^2\theta_W$                   | 30.653   | $\sim 30.6$ | < 0.2% |
| $1/\alpha_s$           | $1/[r^2(1+r-r^2)]$                                    | 8.629    | 8.482       | +1.73% |

All three gauge couplings, and thereby the complete  $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$  boundary condition, are fixed by the single axiom. The Standard Model treats these as four independent experimental inputs.

## 3 One-Loop Running

### 3.1 Beta functions

The SM 1-loop renormalisation-group equations take the form

$$\frac{1}{\alpha_i(\mu)} = \frac{1}{\alpha_i(M_Z)} - \frac{b_i}{2\pi} \ln \frac{\mu}{M_Z}, \quad (11)$$

with SM beta coefficients (no SUSY)

$$b_1 = +\frac{41}{10}, \quad b_2 = -\frac{19}{6}, \quad b_3 = -7. \quad (12)$$

$b_1 > 0$  means the  $\text{U}(1)$  coupling grows with energy;  $b_2, b_3 < 0$  reflect asymptotic freedom in  $\text{SU}(2)$  and  $\text{SU}(3)$ .

### 3.2 Pairwise intersections

Setting  $1/\alpha_i(\mu) = 1/\alpha_j(\mu)$  and solving (??) analytically gives

$$\ln \frac{\mu_{ij}}{M_Z} = \frac{2\pi [1/\alpha_j(M_Z) - 1/\alpha_i(M_Z)]}{b_j - b_i}. \quad (13)$$

With the UM-derived boundary conditions:

| Pair                     | $\mu_{\text{intersect}}$           | $1/\alpha$ at intersection | Third coupling there |
|--------------------------|------------------------------------|----------------------------|----------------------|
| $\alpha_1 \cap \alpha_2$ | $2.686 \times 10^{14} \text{ GeV}$ | 45.12                      | $1/\alpha_3 = 40.62$ |
| $\alpha_1 \cap \alpha_3$ | $3.443 \times 10^{15} \text{ GeV}$ | 35.81                      | $1/\alpha_2 = 24.00$ |
| $\alpha_2 \cap \alpha_3$ | $4.335 \times 10^{17} \text{ GeV}$ | -1.73                      | (unphysical)         |

The pairwise intersections span a band  $\mu \sim 10^{14}\text{--}10^{17}$  GeV. The three couplings do not meet at a single point under SM 1-loop running. This is a well-established result; it does not falsify UM, since UM inherits whatever UV completion is placed above  $M_{\text{GUT}}$ .

### 3.3 Running table

| $\log_{10}(\mu/\text{GeV})$ | $\mu$         | $1/\alpha_1$ | $1/\alpha_2$ | $1/\alpha_3$ |
|-----------------------------|---------------|--------------|--------------|--------------|
| 2                           | 100 GeV       | 59.5         | 29.6         | 9.0          |
| 6                           | $10^6$ GeV    | 50.9         | 23.6         | 13.6         |
| 10                          | $10^{10}$ GeV | 42.2         | 17.6         | 18.2         |
| 14                          | $10^{14}$ GeV | 33.6         | 11.6         | 22.8         |
| 16                          | $10^{16}$ GeV | 29.8         | 8.3          | 24.9         |
| 18                          | $10^{18}$ GeV | 26.1         | 5.0          | 26.9         |

The  $\alpha_1$  and  $\alpha_2$  curves cross near  $10^{14}\text{--}10^{15}$  GeV. The strong coupling overshoots. In the MSSM the beta coefficients shift to  $b_1^{\text{MSSM}} = 33/5$ ,  $b_2^{\text{MSSM}} = 1$ ,  $b_3^{\text{MSSM}} = -3$ , and all three couplings converge near  $2 \times 10^{16}$  GeV. UM makes no independent statement about SUSY; it simply sets the boundary conditions.

## 4 The Unification Scale

### 4.1 UM-natural scales from the Planck mass

The Planck mass  $M_{\text{Pl}} = 1.221 \times 10^{19}$  GeV is the natural ultraviolet scale of any quantum-gravitational framework. In UM, sub-Planck scales arise by contracting  $M_{\text{Pl}}$  through integer powers of  $\varphi^{-1}$ , since each recursion step contracts lengths by  $r = 1/(2\varphi) \sim \varphi^{-1}$  in the relevant channel.

The sequence of UM-natural scales is

$$\mu_n = M_{\text{Pl}} \varphi^{-n}, \quad n = 0, 1, 2, \dots \quad (14)$$

### 4.2 The $\varphi^{-14}$ GUT scale

Powers of  $\varphi$  grow as Fibonacci numbers:  $\varphi^{14} = 843.0$  and  $\varphi^{13} = 521.0$  (exact integer nearest values, consistent with the Lucas–Fibonacci identity). The corresponding sub-Planck scales are

$$M_{\text{GUT}}^{(14)} = M_{\text{Pl}} \varphi^{-14} = \frac{1.221 \times 10^{19}}{843.0} = 1.448 \times 10^{16} \text{ GeV}, \quad (15)$$

$$M_{\text{GUT}}^{(13)} = M_{\text{Pl}} \varphi^{-13} = \frac{1.221 \times 10^{19}}{521.0} = 2.344 \times 10^{16} \text{ GeV}. \quad (16)$$

Both values lie within the pairwise-intersection band identified in Section 3. The canonical MSSM unification scale is  $M_{\text{GUT}}^{\text{MSSM}} \approx 2 \times 10^{16} \text{ GeV}$ , consistent with both  $\mu_{13}$  and  $\mu_{14}$ . We take  $\mu_{14}$  as the primary UM value because it lies closer to the  $\alpha_1 \cap \alpha_3$  intersection and because  $14 = 2 \times 7$  corresponds to the second-generation colour depth in the recursion.

### 4.3 Physical interpretation

The UM framework derives  $M_{\text{GUT}} \approx 1.45 \times 10^{16} \text{ GeV}$  without any GUT model input. The scale emerges from the ratio of the Planck mass to the 14th power of the golden ratio, a consequence of counting recursion cycles from the Planck scale to the weak scale. Fourteen cycles traverse: four colour-sector traversals, two electroweak-sector traversals, and eight compactification cycles of the  $G_2$  manifold.

The quantitative interpretation of these cycle counts is the subject of the companion heterotic paper. Here we note only that the pairwise intersection band  $10^{14}$ – $10^{17} \text{ GeV}$  and the UM-natural scale  $M_{\text{Pl}} \varphi^{-14}$  are consistent at the level of the width of the band.

## 5 Prediction Table

| Observable             | UM expression                                         | UM value                          | Observed                            | Error       |
|------------------------|-------------------------------------------------------|-----------------------------------|-------------------------------------|-------------|
| $1/\alpha_{\text{em}}$ | $\varphi^{10} + \varphi^5 + \varphi^2 + \varphi^{-2}$ | 137.082                           | 137.036                             | +0.03%      |
| $\sin^2 \theta_W$      | $r/[2(1-r)]$                                          | 0.22361                           | 0.22290                             | +0.32%      |
| $1/\alpha_2$           | $\alpha_{\text{em}}/\sin^2 \theta_W$                  | 30.653                            | $\sim 30.6$                         | —           |
| $\alpha_3(M_Z)$        | $r^2(1+r-r^2)$                                        | 0.11591                           | 0.11790                             | −1.69%      |
| $M_{\text{GUT}}$       | $M_{\text{Pl}} \varphi^{-14}$                         | $1.45 \times 10^{16} \text{ GeV}$ | $\sim 2 \times 10^{16} \text{ GeV}$ | within band |

Every entry in the “Observed” column is an independent experimental input in the Standard Model. UM derives them all from the axiom  $c^2 = c + 1$  with zero free parameters.

## 6 Discussion

### 6.1 Significance of fixing the boundary condition

The claim that UM “fixes all three gauge couplings” requires care. The Standard Model uses the measured values of  $\alpha_{\text{em}}$ ,  $G_F$  (Fermi constant), and  $\alpha_s$  as experimental inputs, and from them derives all electroweak observables. UM replaces those three inputs with the single real number  $r = 1/(2\varphi)$ , itself derived from the axiom.

This represents a reduction from three free parameters to zero across the gauge sector. The residuals (0.03%, 0.32%, 1.69%) are all sub-floor, meaning no additional tuning cycle is geometrically required.

### 6.2 Non-unification at 1-loop is inherited, not fatal

It has been known since the 1990s that SM gauge couplings do not meet at a single point under 1-loop running [?]. SUSY threshold corrections shift the beta coefficients and restore 3-way unification at  $\sim 2 \times 10^{16}$  GeV. UM does not independently derive whether the SM or MSSM beta functions govern the running above  $M_Z$ ; that depends on the spectrum of particles in the  $10^2$ – $10^3$  GeV range, which is determined by the second  $E_8$  sector of the UM heterotic construction.

The result here is therefore: *UM derives the boundary conditions; the running then proceeds as in whichever extension of the SM is correct.* If SUSY particles exist near the TeV scale, 3-way unification follows from the UM boundary conditions at  $\sim 2 \times 10^{16}$  GeV.

### 6.3 The GUT scale as a Fibonacci sub-Planck scale

The appearance of  $\varphi^{14}$  is not a fit. The sequence  $M_{\text{Pl}}\varphi^{-n}$  is the only dimensionless family of mass scales constructible from  $M_{\text{Pl}}$  and the UM axiom. The fact that  $n = 14$  falls within the observed unification band is a non-trivial consistency check: the axiom produces a GUT-scale hierarchy without introducing a new large integer.

In string units,  $M_s \sim M_{\text{Pl}}/\sqrt{\alpha'}$  and threshold corrections introduce further  $\varphi$  suppressions; a more complete treatment is left to the companion heterotic paper.

### 6.4 What remains

- **MSSM threshold corrections.** Including SUSY partners near 1–10 TeV shifts the beta coefficients and may produce exact 3-way unification at the UM-derived  $M_{\text{GUT}}$ .

- **Proton decay rate.**  $M_{\text{GUT}} \approx 1.45 \times 10^{16} \text{ GeV}$  implies a proton lifetime  $\tau_p \sim M_{\text{GUT}}^4/m_p^5 \sim 10^{35} \text{ yr}$ , testable at Hyper-Kamiokande.
- **Heterotic  $E_8 \times E_8$  companion.** The second  $E_8$  sector provides the UV completion that enforces full 3-way unification. Its derivation is the subject of a forthcoming paper.

## Closing

One axiom. Three gauge couplings. Zero free parameters.

The Standard Model’s gauge sector requires three independently measured inputs. UM derives all three from  $c^2 = c + 1$ , places the GUT scale at  $M_{\text{Pl}}\varphi^{-14}$ , and leaves the running to whichever UV completion is correct. The geometry does the work; the physics follows.

## References

- [1] U. Amaldi, W. de Boer, H. Fürstenau, *Comparison of grand unified theories with electroweak and strong coupling constants measured at LEP*, Phys. Lett. B **260** (1991) 447.
- [2] R. L. Workman *et al.* (Particle Data Group), *Review of Particle Physics*, Prog. Theor. Exp. Phys. **2022**, 083C01 (2022).